

Causal discovery for survey data?

Preliminary data

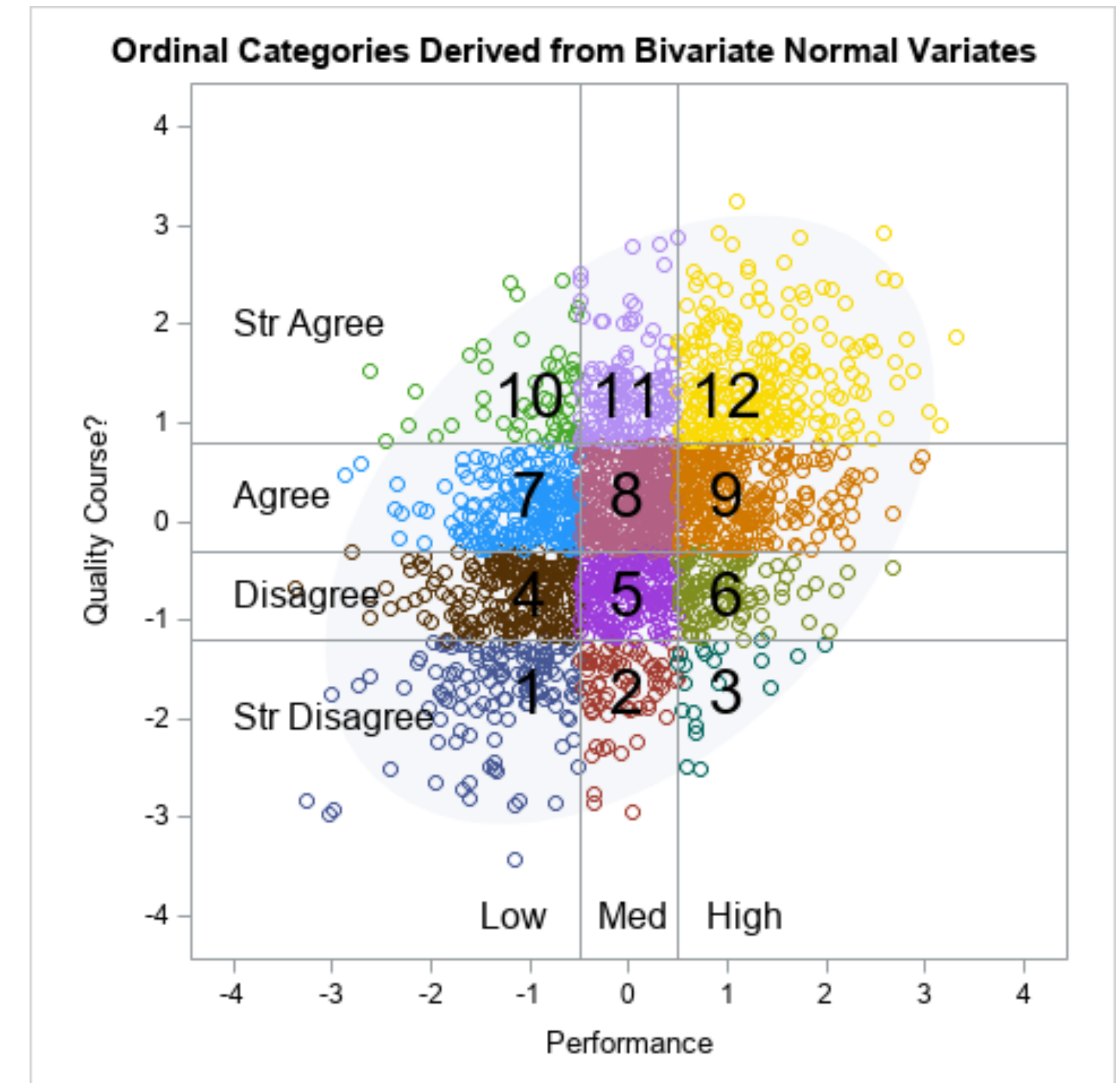
SHEP-SDOH team meeting

2025-11-20

Step 1: Polychoric Correlation & Clustering

Polychoric correlation: why

- Survey data is ordinal: must use polychoric correlation.
- In polychoric correlation:
 - Find 'good' latent distribution (scatterplot of points)
 - Cut x and y axis of scatterplot into different groups, gives us categorical data.
 - Estimate correlation (Pearson) of the 'good' latent distribution.
 - Applicable for non-ordinal categorical data too.
- Scale of polychoric correlation is -1 to +1.
- **We consider thirteen needs (3 levels each)**
- **We consider two health/wellbeing variables:**
 - **For wellbeing score (scale of 0-100), cut into quartiles, Q4 being best, Q1 being worst.**
 - **For mental health (5 levels, from Excellent to Poor).**

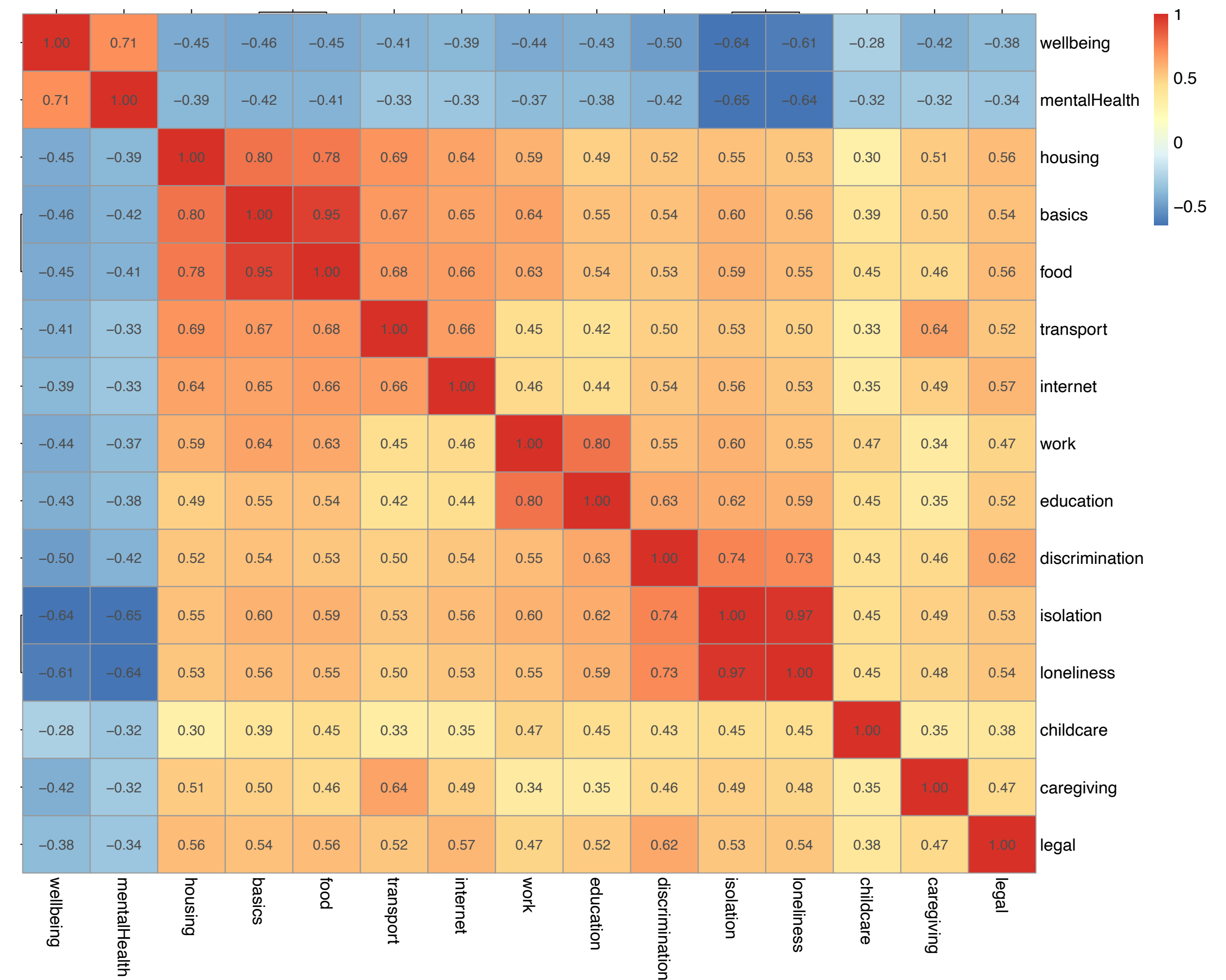


Not from SHEP/SDOH data, for explanation of polychoric correlation

Step 1: Polychoric Correlation & Clustering

Polychoric correlation: why

- Wellbeing (WB) and Mental Health (MH) are positively correlated with each other (0.70)
- All social needs are positively correlated (spread between 0.30 – 0.80).
- WB and MH are negatively correlated with needs (people reporting needs have poorer WB and MH outcomes)
- WB and MH report strongest (negative) correlation with isolation and loneliness (~ -0.60), followed by discrimination (~ -0.50).



Outcomes vs needs (ordering unclear)

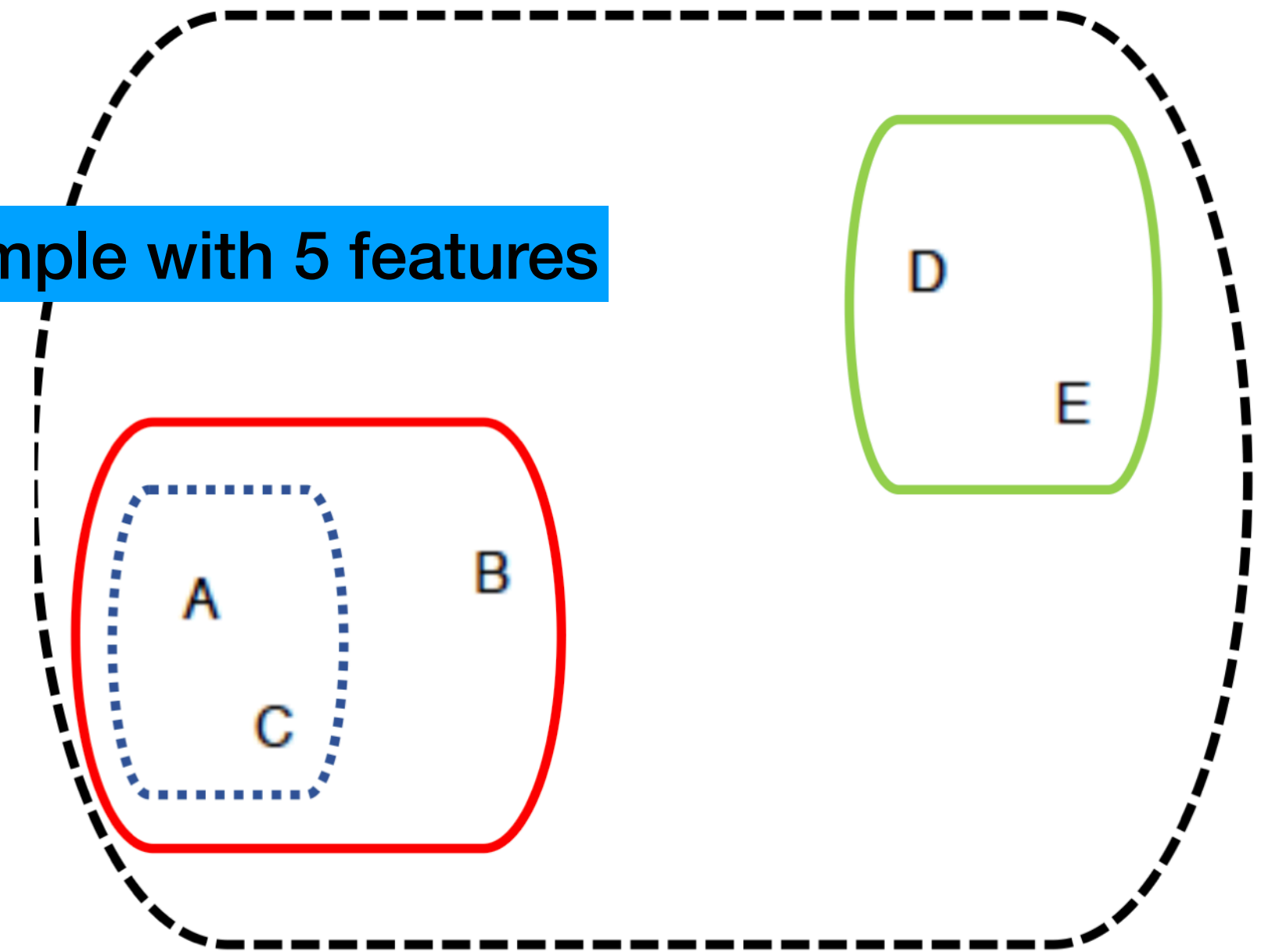
There are clusters of needs (coming to this in next slide).

Hierarchical clustering

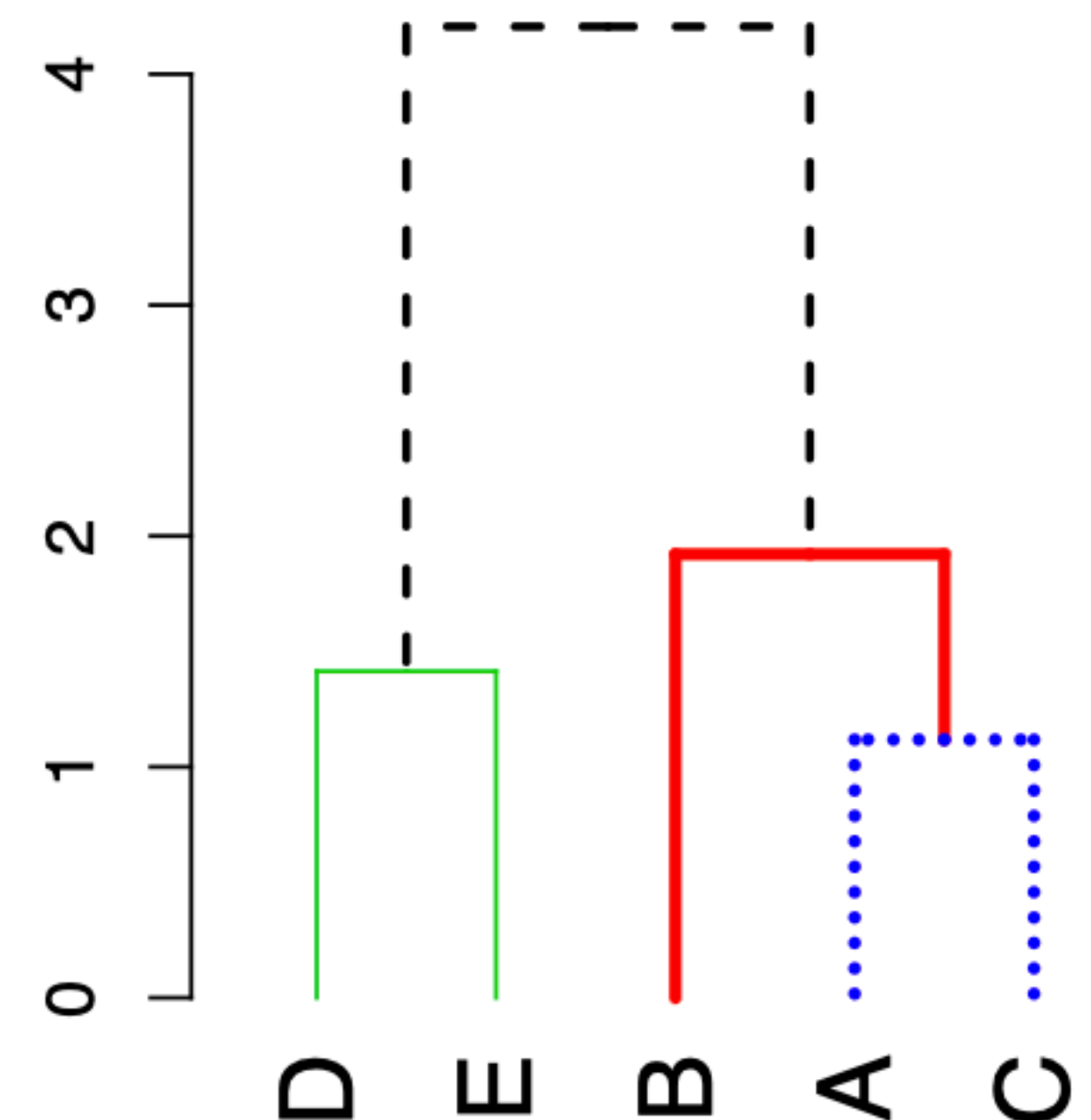
Grouping “similar” features together

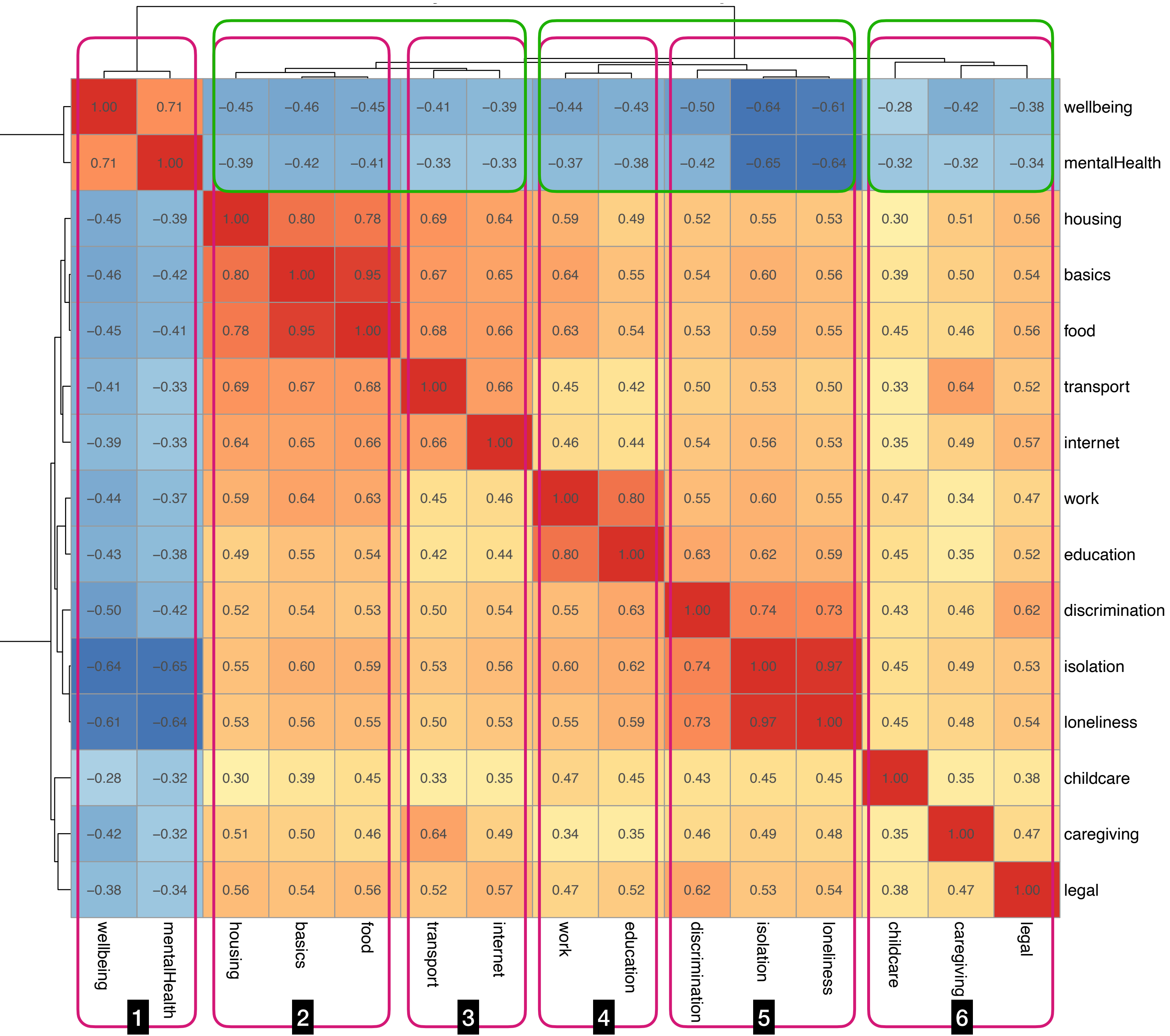
- Begin with p observations and measure all $p(p - 1)/2$ **inter-feature distances**. Treat each feature as it's own island.
- For $i = p, p - 1, \dots, 2$:
 - Examine all pairwise **inter-cluster dissimilarities** among the i clusters and identify the pair of clusters that are **most similar** and fuse them.
 - **Dissimilarity** between two clusters indicates the height at which in the dendrogram at which the fusion is placed.
 - Compute new pairwise **inter-cluster dissimilarities** among the $i - 1$ remaining clusters. Repeat step above.

Example with 5 features



Dendrogram





Interpreting the clustering scheme: The tree-like diagram on the left and top, called a dendrogram, is a visual tool that groups similar variables together. Think of it like a family tree: items that are "related" (highly correlated) are joined by branches that are close to the heatmap.

Short branches that merge "low" (very close to the heatmap) connect items that are extremely similar, like loneliness and isolation (which are almost identical). Long branches that merge "high" (far to the left, like the main trunk) connect groups that are very different from each other.

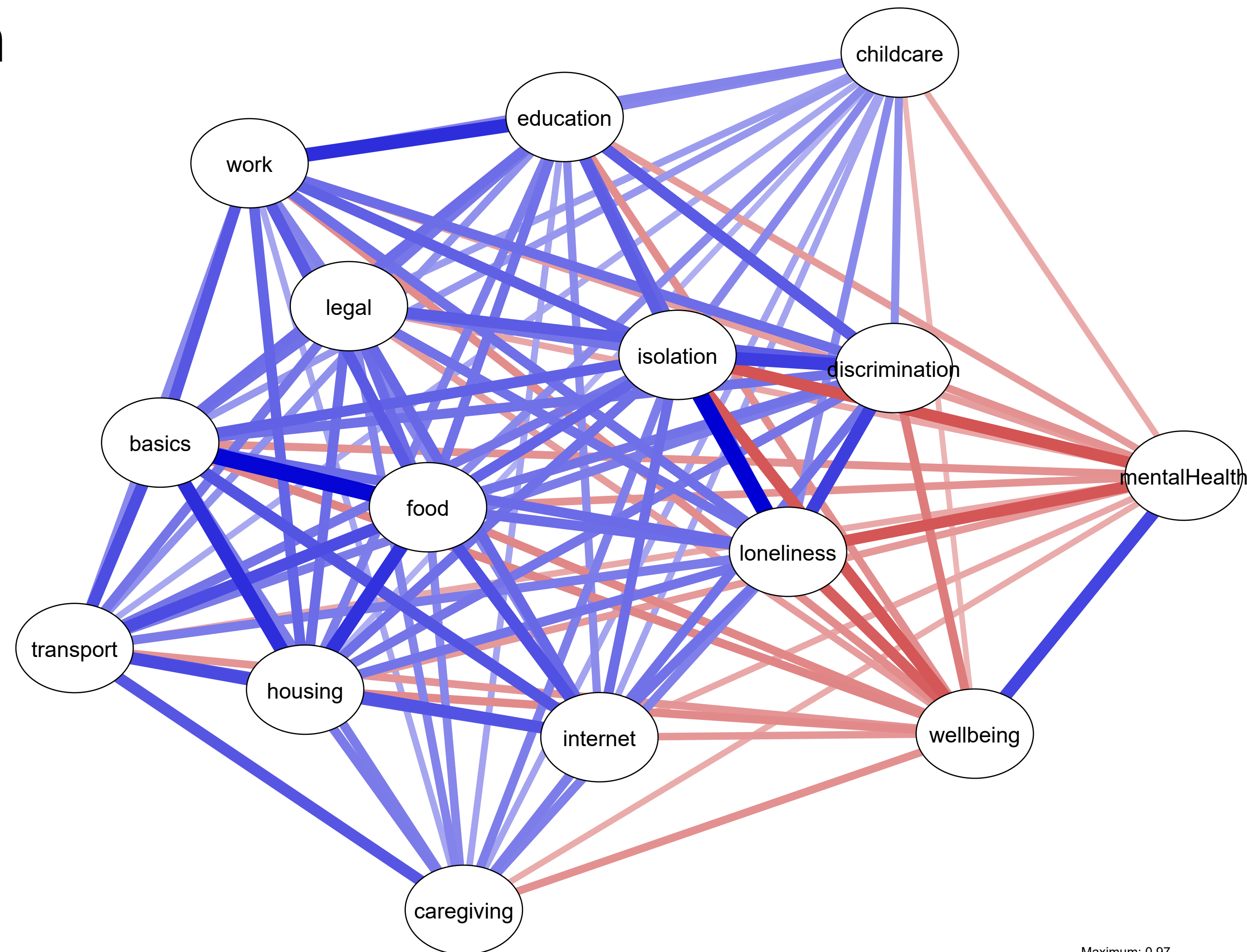
- Just by looking at strength of correlations: we get six clusters.
- Cluster 1: WB/MH are strongly and positively correlated with each other, and strongly negatively correlated (all blue cells) with every cluster.
- C2: Basic needs
- C3: Connectivity (real + virtual)
- C4: Socioeconomic (better word?)
- C5: Psychosocial (better word?)
- C6: Household (better word?)
- C1 and C2-C6 form distinct groups.

Outcomes vs needs (ordering unclear)

Step 2A: Correlation network

Undirected (and dense) graph

- From the correlation matrix, build a network.
- Each node is a need/WB/MH.
- Lines joining any two nodes are weighted, stronger correlations mean thicker lines.
- Positive associations are shaded blue, negative associations are shaded red.

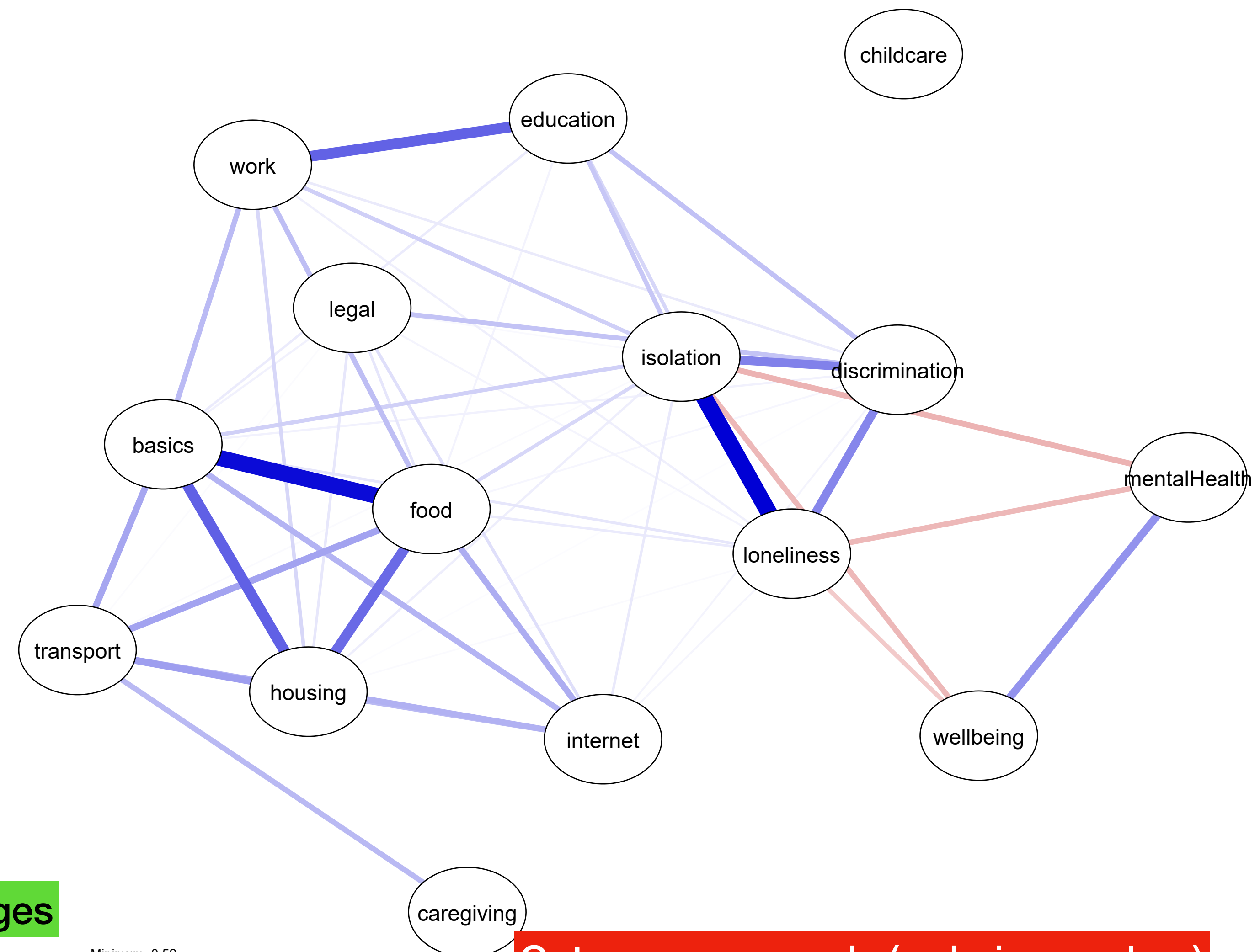


Outcomes vs needs (ordering unclear)

Step 2B: **TRIMMED** Correlation network

Undirected (and sparse) graph

- From the correlation matrix, build a network.
- Each node is a need/WB/MH.
- Lines joining any two nodes are weighted, stronger correlations mean thicker lines.
- Positive associations are shaded blue, negative associations are shaded red.
 - Trimming 'lighter' edges = removing 'weaker' associations
 - Post-trimming, the clusters from the heatmap become clearer.



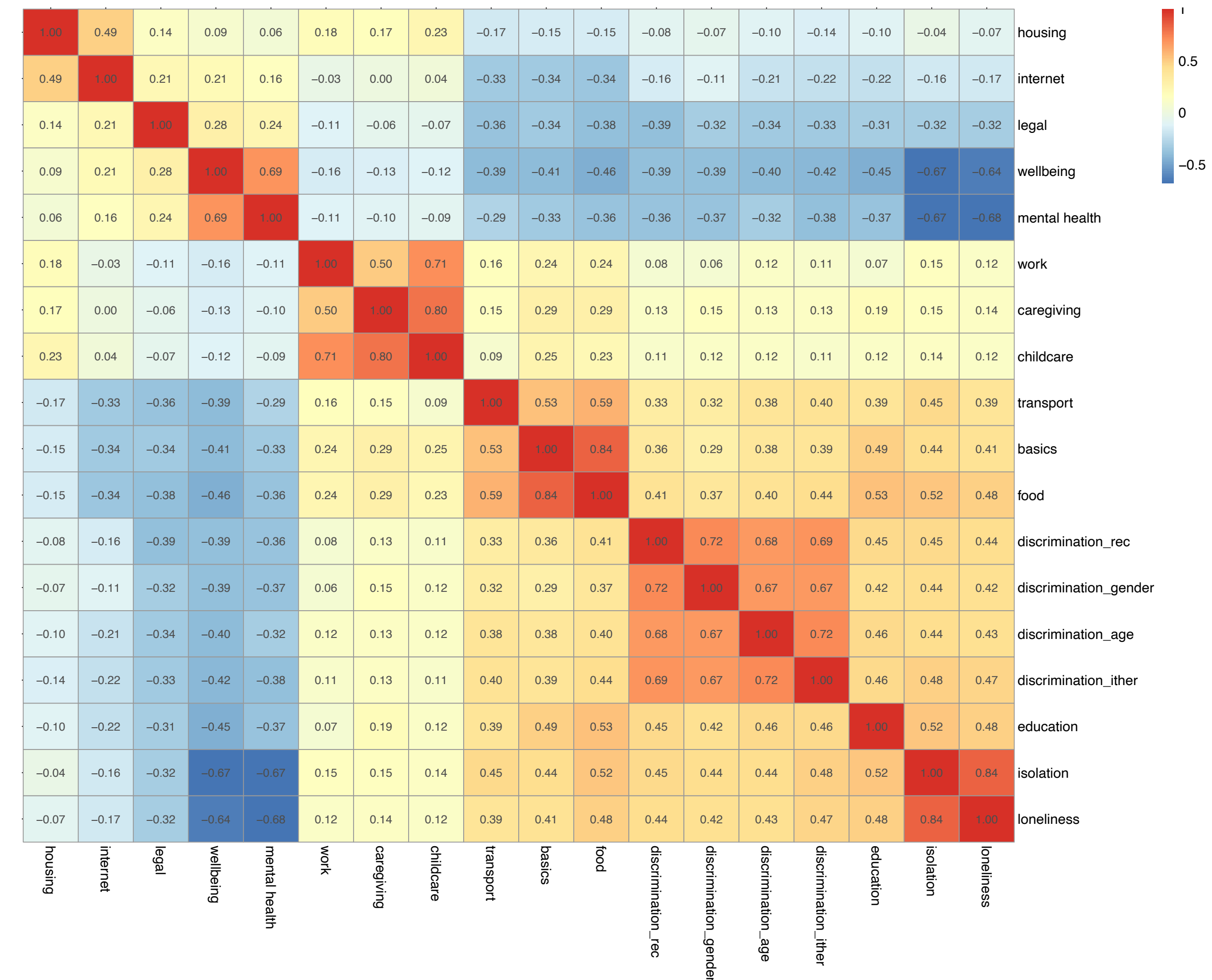
Next time: we will assign direction (arrows) to these edges

Outcomes vs needs (ordering unclear)

Step 1: Polychoric Correlation & Clustering

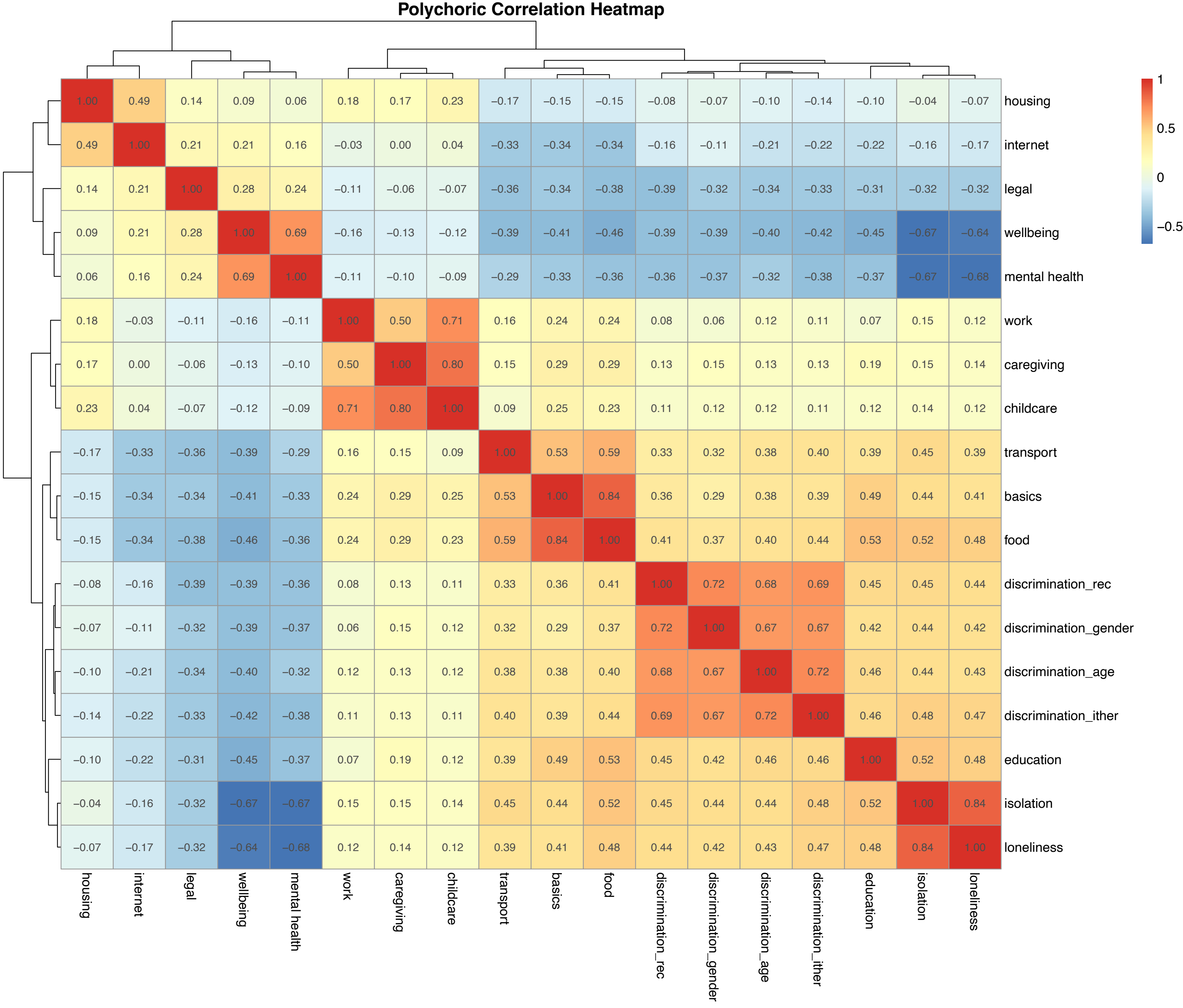
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Outcomes vs risks (ordering clear)

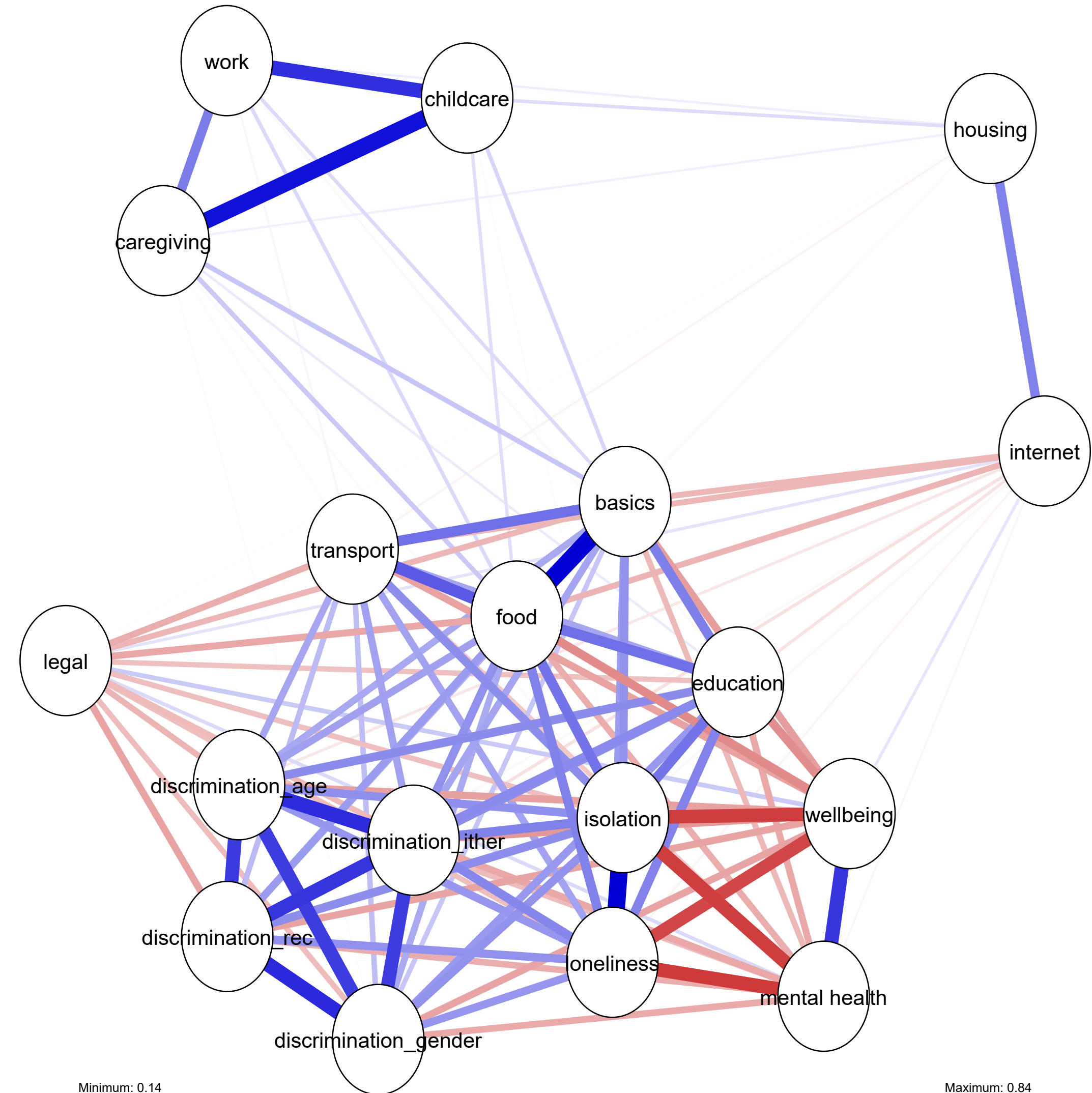


Outcomes vs risks (ordering clear)

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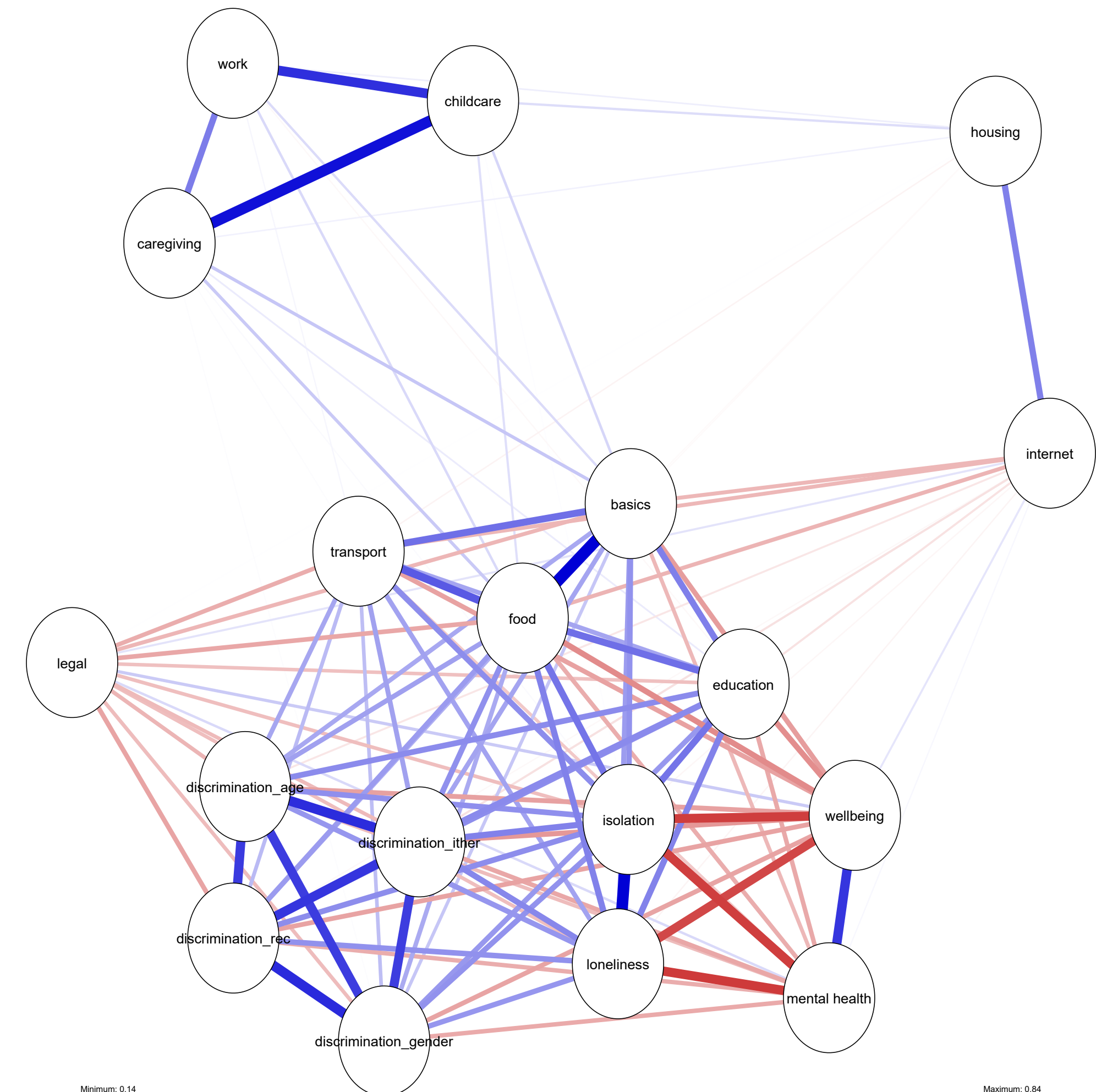


Outcomes vs risks (ordering clear)

Step 2B: **TRIMMED** Correlation network

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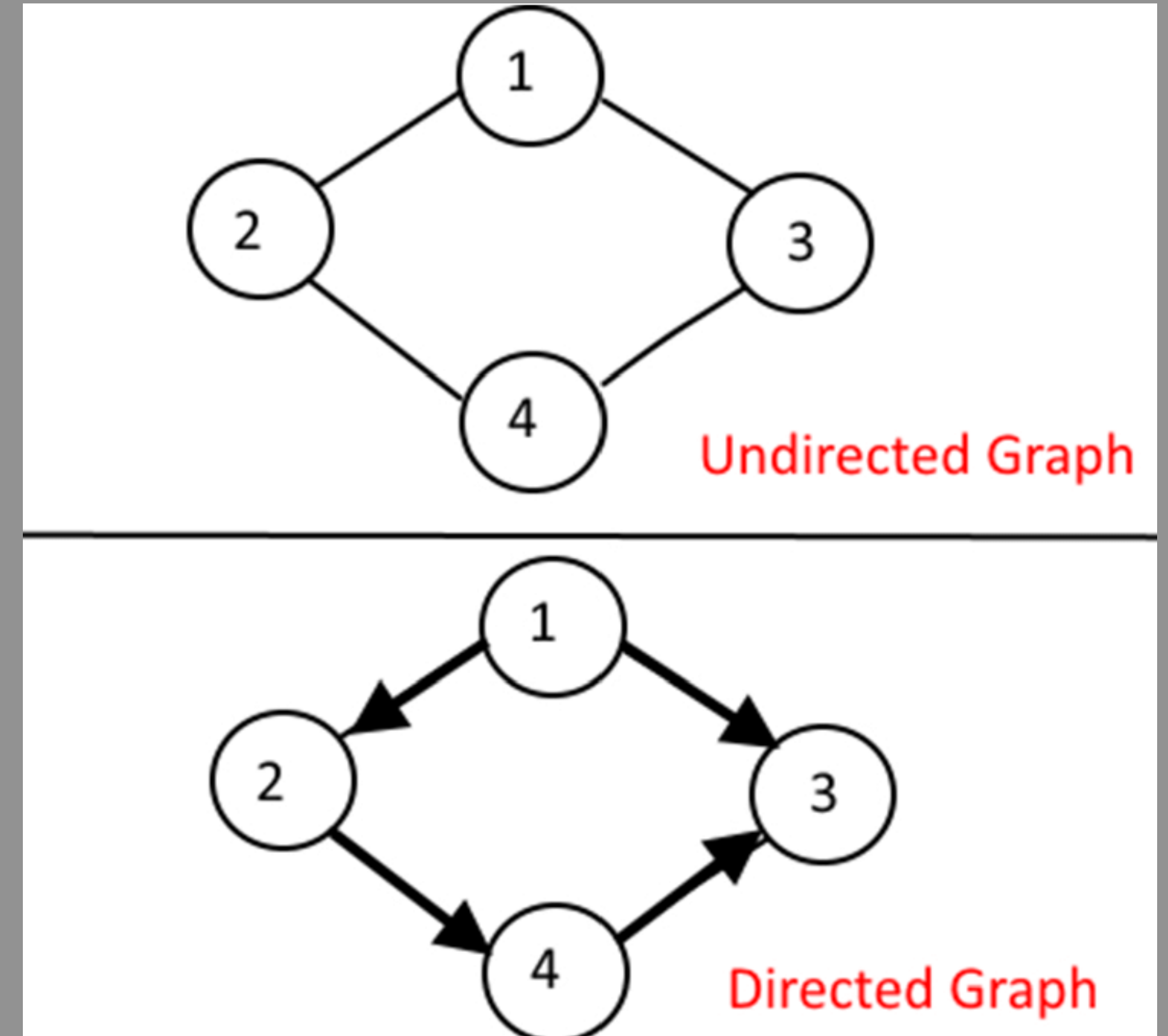
2025-10-23

SLIDES FROM PREVIOUS MEETING (FOR CONTINUITY)

DAG: Directed acyclic graphs

Tool for Causal Thinking

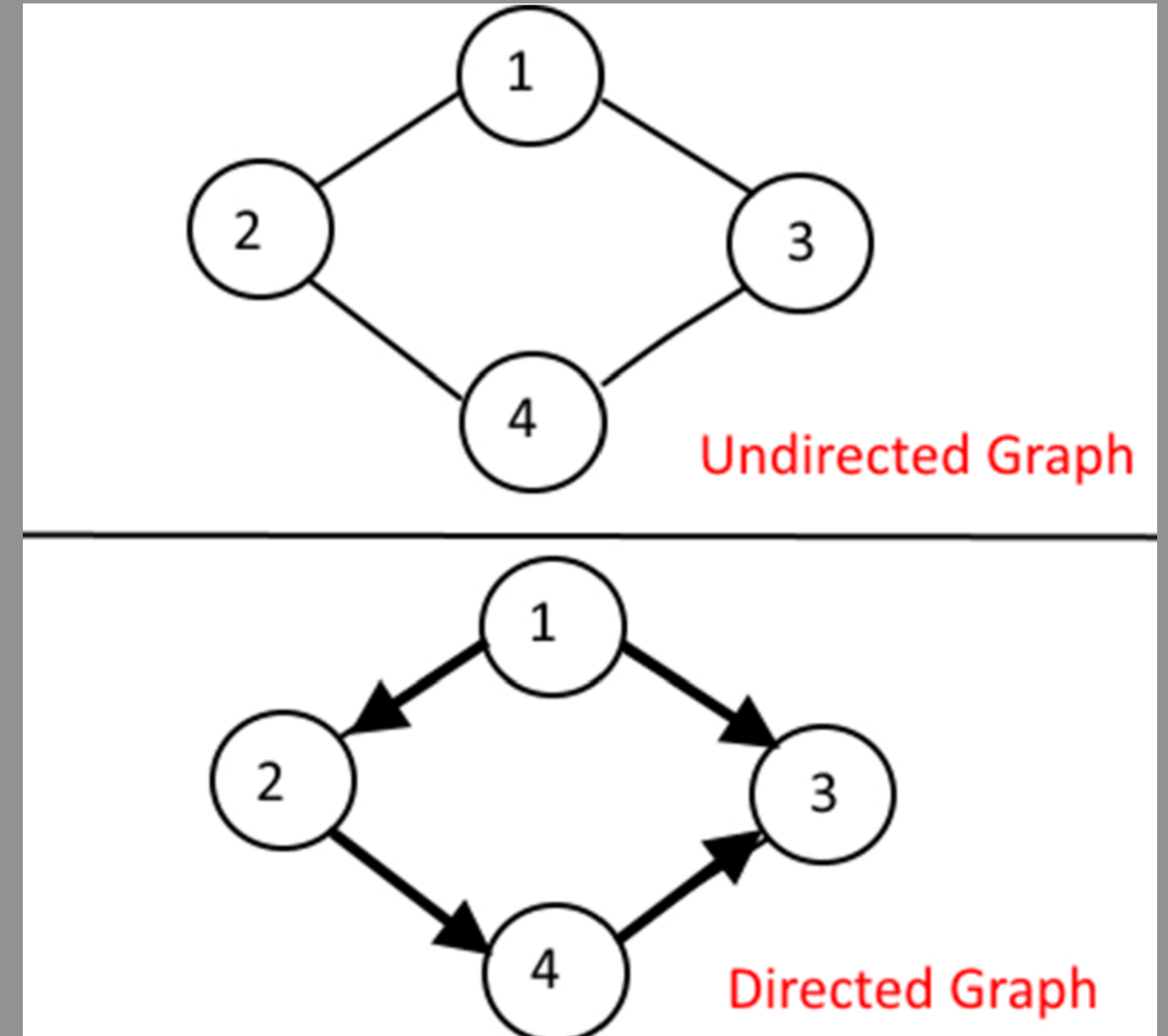
- It's a visual map of our causal assumptions about the world.
- Nodes = Variables (e.g., "Smoking," "Patient Satisfaction")
- Arrows = Causal paths (e.g., Smoking → Lung Cancer)
- Lines = Association paths (smoking and lung cancer are associated)
- They are a cornerstone of modern epidemiology and causal modeling.
- Formulate hypotheses and identify crucial problems like confounding and selection bias.



Correlation vs. Causation

Insights from graphs

- Undirected Graphs or Correlation Networks.
 - What they are: Nodes connected by lines (edges), **not arrows**.
 - What they show: The strength of an association.
 - They are ambiguous. **Does Loneliness lead to Poor Health, or does Poor Health lead to Loneliness? This graph can't tell them apart.**
- Directed Graphs (Causal Networks)
 - What they are: Nodes connected by arrows.
 - What they show: A specific causal pathway.
 - Example: Loneliness → Poor Health. **This is what we need to design effective interventions.**



Correlation Map for SHEP data

Starting point

- Calculate a large correlation matrix for (all/some) of the questions (e.g., Provider Communication, Overall Rating, Social Circumstances).
- Visualize this as an undirected "network graph."
- What this tells us: Which patient experiences are most strongly related.
 - Expect to see see a strong link between "Provider Listening" and "Overall Rating"?
- What it doesn't tell us: **Does listening cause the high rating? Or are both caused by a third factor (e.g., an easy-to-treat patient)?**

Causal DAGs

Moving beyond heatmaps

- These are algorithms that can learn the **most likely** causal structure (the DAG) directly from the data.
- OrdCD method (Ni et al. 2025): specifically designed for ordinal survey data (like the "Never" to "Always" scales in SHEP).
- It's been proven to be able to identify the correct causal direction (e.g., $A \rightarrow B$ or $B \rightarrow A$) where other methods fail.
- This lets us move from a simple correlation map to a plausible causal DAG of the patient experience.

Potential projects

Tailoring OrdCD for SHEP data

- Extension 1: Creating a Statistically "True" Causal Map
 - Problem: The current model assumes a simple random sample. Our SHEP data uses complex survey weights to be representative.
 - Project: adapt the OrdCD algorithm to **correctly incorporate survey weights**.
 - Deliverable: statistically robust and generalizable (to VA superpopulation) findings.
- Extension 2: Causal Discovery in Diverse Populations
 - Problem: The current model assumes one causal map for everyone.
 - Project: extend the OrdCD model to **explore heterogeneity**. We'll allow the causal links to change based on demographic covariates (age, race-ethnicity, sex).
 - Deliverable: "Is the causal link from 'Provider Listening' to 'Patient Trust' stronger for minority patients than for white patients?"